QBITEL BRIDGE

Telecommunications & 5G Networks

DEPLOYMENT
CHECKLIST

Phases 0-10: Pre-Engagement to Go-Live

SS7 Signaling | 5G Core | IoT/mMTC | Fraud | Compliance

11

Phases

0	1	2	3	4	5	6	7	8	9	10
PRE	DISC	INFRA	POL	SIG	5G	FRAUD	IOT	MON	COMP	GO

Deployment Overview

This checklist guides the end-to-end deployment of QBITEL Bridge across a telecommunications operator network -- from initial pre-engagement assessment through to full go-live and operational handover. Each phase includes specific technical items, deliverables, and RACI assignments. Items marked with * are critical path -- they must be completed before the next phase begins.

*** Critical items are marked in gold and must be signed off before phase gate.**

PHASE	NAME	DURATION	OWNER	ITEMS
0	**Pre-Engagement**	1 week	QBITEL SE + Operator CISO	8
1	**Protocol Discovery**	1-2 weeks	QBITEL SE + Network Ops	9
2	**Infrastructure Readiness**	1-2 weeks	Operator Infra + QBITEL	8
3	**Security Policy Configuration**	1 week	Operator Security + QBITEL	8
4	**Signaling Protection**	2-3 weeks	Signaling Ops + QBITEL	9
5	**5G Core Security**	2 weeks	Core Network Ops + QBITEL	9
6	**Fraud Detection Activation**	1-2 weeks	Fraud Team + QBITEL	8
7	**IoT Gateway Protection**	1-2 weeks	IoT/Data Team + QBITEL	7
8	**Monitoring & Alerting**	1 week	NOC + QBITEL	7
9	**Compliance Validation**	1 week	Compliance + Legal	8
10	**Go-Live & Handover**	3-5 days	Joint QBITEL + Operator	6

PHASE 0

PRE-ENGAGEMENT

1 week | QBITEL SE + Operator CISO

The pre-engagement phase establishes the scope, objectives, and success criteria for the QBITEL Bridge deployment. No network changes occur in this phase. Output is a signed Statement of Work and a completed Threat Exposure Assessment Report.

- [ ] * **Complete network topology questionnaire -- RAN, core, interconnect, roaming, and cloud inventory** — PLANNING
- [ ] * **Identify all SS7 interconnect points -- E1/T1 and SIGTRAN/M3UA links to roaming partners and IPX hubs** — SS7
- [ ] Document existing SS7 signaling firewall vendor, version, and current rule policy scope — SS7
- [ ] * **Map 5G deployment status -- NSA vs SA, NFs deployed, slices active, vendor ecosystem** — 5G
- [ ] * **Identify regulatory obligations -- NIS2, FCC reporting, GSMA FS.19 assessment deadlines** — COMPLIANCE
- [ ] Obtain fraud loss profile -- IRSF, SIM swap, bypass fraud -- 12 months of historical data — FRAUD
- [ ] Define VIP subscriber categories requiring enhanced SS7 location protection — SS7
- [ ] * **Execute NDA and initiate Statement of Work review and approval** — ADMIN

PHASE DELIVERABLES

- Signed NDA and Statement of Work
- Network Topology Map (RAN, Core, Interconnect)
- Threat Exposure Assessment Report (confidential)
- Regulatory Obligation Register
- Fraud Loss Profile -- 12 month baseline

PHASE 1

PROTOCOL DISCOVERY

1-2 weeks | QBITEL SE + Network Ops

Passive protocol discovery establishes a complete inventory of signaling traffic, protocol versions, and current security posture without any changes to live network traffic flows. QBITEL Bridge passive taps are installed and configured for monitoring-only mode.

- [ ] * **Install QBITEL Bridge passive taps on SS7/SIGTRAN links -- monitoring only, no inline enforcement** — SS7
- [ ] * **Run 72-hour SS7 MAP traffic capture and baseline analysis -- document normal roaming patterns** — SS7
- [ ] Enumerate all SS7 MAP message types in active use -- SRI, ATI, PSI, UL, CLR, ISD — SS7
- [ ] * **Capture and analyze Diameter S6a/S6d/S9 traffic -- identify authentication vector exchange patterns** — DIAMETER
- [ ] * **Map 5G SBI interface traffic -- identify all NF-to-NF API call patterns and volumes** — 5G
- [ ] Enumerate active network slices (S-NSSAI) and document per-slice traffic profiles — 5G
- [ ] Capture SIP trunk traffic -- identify IRSF-risk destinations and anomalous call patterns — SIP
- [ ] Document all IoT APNs, NB-IoT/LTE-M device populations, and current security policies — IOT
- [ ] * **Generate Protocol Inventory Report with quantum vulnerability risk scoring per interface** — COMPLIANCE

PHASE DELIVERABLES

- SS7 MAP Traffic Baseline Report
- Protocol Inventory -- all interfaces with quantum vulnerability scores
- Active Attack Campaign Report (detected during passive monitoring)
- 5G Slice Inventory and Traffic Profile
- IoT APN and Device Population Report

PHASE 2

INFRASTRUCTURE READINESS

1-2 weeks | **Operator Infra + QBITEL**

Infrastructure readiness prepares the hardware, HSM, network integration, and software environment for QBITEL Bridge deployment. No security policy changes are active at this stage.

- [ ] * **Provision QBITEL Bridge server hardware -- 2-socket x86 with AVX-512, minimum 64GB RAM per node** — HARDWARE
- [ ] * **Install and initialize Hardware Security Modules (HSM) -- Thales Luna or Entrust nShield** — HSM
- [ ] * **Perform HSM key ceremony -- establish PQC root key hierarchy under dual-control procedures** — HSM
- [ ] * **Configure HSM cluster replication for geographic redundancy -- minimum 2 sites** — HSM
- [ ] Deploy QBITEL Bridge software -- containerized (Kubernetes) or bare-metal per architecture decision — DEPLOY
- [ ] Configure network taps / inline deployment ports for SS7 and SIP interfaces — NETWORK
- [ ] Validate network management connectivity -- NETCONF/YANG for Nokia, RESTCONF for Ericsson — INTEGRATION
- [ ] * **Configure QBITEL Bridge management console access -- RBAC, MFA, audit logging** — SECURITY

PHASE DELIVERABLES

- HSM Installation and Key Ceremony Report
- QBITEL Bridge Cluster Deployment Verification
- Network Integration Test Report
- Management Console Access and RBAC Configuration
- Infrastructure Readiness Sign-off (joint QBITEL + Operator)

PHASE 3

SECURITY POLICY CONFIGURATION

1 week | Operator Security + QBITEL

Security policy configuration establishes the cryptographic policy framework, subscriber data classification, and protocol-specific security rules before any enforcement is activated. All policies are reviewed and approved by the operator security team before phase gate.

- [ ] * **Define subscriber data classification tiers -- standard, VIP, critical government, MVNO** — POLICY
- [ ] * **Configure PQC algorithm policy per network domain -- choose ML-KEM-768 vs 1024 per interface** — CRYPTO
- [ ] * **Define SS7 MAP filtering policy -- alert-only vs inline block, per attack category** — SS7
- [ ] * **Configure Diameter S6a roaming partner trust profiles -- whitelist legitimate partners** — DIAMETER
- [ ] * **Define 5G network slice security policies -- per S-NSSAI PQC key hierarchy and isolation rules** — 5G
- [ ] Configure SIP trunk IRSF blocking policy -- thresholds, automatic trunk suspension rules — SIP
- [ ] Define IoT security profiles per APN -- NB-IoT, LTE-M, industrial IIoT enhanced profiles — IOT
- [ ] Configure fraud detection sensitivity -- precision vs recall tradeoff per fraud category — FRAUD

PHASE DELIVERABLES

- Security Policy Framework Document (operator-approved)
- PQC Algorithm Selection Matrix per Network Domain
- SS7/Diameter Filtering Policy Rulebook
- 5G Slice Security Policy Configuration
- Fraud Detection Sensitivity Configuration

PHASE 4

SIGNALING PROTECTION

2-3 weeks | Signaling Ops + QBITEL

Signaling protection activates inline SS7 MAP filtering, Diameter security, and SIP trunk hardening in production. Activation follows a phased approach: alert-only for 72 hours, graduated enforcement, then full blocking on confirmed attack patterns.

- [ ] * **Activate SS7 MAP AnyTimeInterrogation (ATI) filtering -- alert-only mode for 72 hours** SS7
- [ ] * **Review ATI alert queue -- validate detection accuracy, tune false positive threshold** SS7
- [ ] * **Enable SS7 MAP ATI inline blocking for high-confidence attack signatures** SS7
- [ ] Activate SS7 MAP SendRoutingInfo (SRI) anomaly detection -- roaming partner behavior profiling SS7
- [ ] Enable SS7 MAP ProvideSubscriberInfo (PSI) and UpdateLocation filtering SS7
- [ ] * **Activate PQC authentication vector wrapping on HLR/HSS MAP interfaces** SS7
- [ ] Configure Diameter S6a Cancel Location Request (CLR) anomaly detection DIAMETER
- [ ] Enable Diameter Insert Subscriber Data (ISD) integrity verification DIAMETER
- [ ] * **Activate SIP trunk IRSF detection -- alert-only for 48 hours before enabling blocking** SIP

PHASE DELIVERABLES

- SS7 MAP Filtering Activation Report -- 72hr alert-only results
- False Positive Analysis and Threshold Tuning Report
- Inline Blocking Activation Sign-off
- PQC Authentication Vector Wrapping Verification
- Diameter Security Activation Report

PHASE 5

5G CORE SECURITY

2 weeks | Core Network Ops + QBITE

5G core security activation deploys PQC-hybrid mTLS on all SBI interfaces, enforces slice cryptographic isolation, and activates NF integrity verification across AMF, SMF, UPF, AUSF, UDM, and NRF. Deployment follows one NF at a time to minimize risk.

- [ ] * **Deploy QBITEL Bridge sidecar proxies on AMF pods -- verify PQC hybrid mTLS on N1/N2** 5G
- [ ] * **Activate SUPI/SUCI integrity verification on AMF -- alert on unencrypted SUPI exposure** 5G
- [ ] * **Deploy sidecar proxies on SMF -- activate N4/PFCP session binding integrity verification** 5G
- [ ] Enable GTP-TEID integrity monitoring on UPF -- detect tunnel redirect attempts 5G
- [ ] * **Deploy sidecar on AUSF -- activate PQC wrapping of HXRES* authentication vectors** 5G
- [ ] * **Enable UDM subscriber credential quantum-hardening -- encrypt UDR subscriber data at rest** 5G
- [ ] Activate NRF service registration integrity -- detect rogue NF registration attempts 5G
- [ ] * **Configure per-slice PQC key hierarchies -- validate slice isolation enforcement** 5G
- [ ] Verify SEPP N32 interface PQC hybrid key exchange for inter-PLMN roaming security 5G

PHASE DELIVERABLES

- 5G NF Sidecar Deployment Verification Report
- SBI mTLS PQC Key Exchange Validation
- Slice Isolation Test Results (cross-slice attack simulation)
- SUPI Protection Audit Report
- UDM/UDR Quantum-Hardening Verification

PHASE 6

FRAUD DETECTION ACTIVATION

1-2 weeks | Fraud Team + QBITEL

Fraud detection activation initializes machine learning models with operator-specific CDR history, configures real-time IRSF blocking, and integrates with existing fraud management systems. Models are validated against known fraud events before blocking is enabled.

- [ ] * **Load 6-12 months of historical CDR data for ML model training and calibration** — FRAUD
- [ ] * **Validate ML model accuracy against known historical IRSF events -- target 99.9%+ detection rate** — FRAUD
- [ ] * **Configure IRSF number range intelligence feeds -- GSMA, CFCA, operator-specific IPRN lists** — FRAUD
- [ ] * **Activate real-time IRSF call blocking -- test on low-risk SIP trunks first** — FRAUD
- [ ] Enable wangiri one-ring fraud detection and automatic callback blocking — FRAUD
- [ ] Configure SIM swap anomaly detection -- HLR/HSS update pattern monitoring — FRAUD
- [ ] Enable bypass fraud / SIM box detection -- RF signature and call pattern analysis — FRAUD
- [ ] * **Integrate fraud event feed with existing FMS (Subex/TEOCO/Syniverse) via REST API** — INTEGRATION

PHASE DELIVERABLES

- ML Model Training and Validation Report
- IRSF Detection Accuracy Benchmark Results
- Real-time Blocking Activation Report
- FMS Integration Test Confirmation
- First Month Fraud Prevention Impact Report

PHASE 7

IOT GATEWAY PROTECTION

1-2 weeks | IoT/Data Team + QBITEL

IoT gateway protection activates network-layer PQC for NB-IoT, LTE-M, and eMTC device populations without requiring any firmware updates to constrained devices. Botnet detection and automatic quarantine are configured per device category.

- [ ] * **Configure IoT PQC gateway per NB-IoT and LTE-M APN -- verify device traffic forwarding** IOT
- [ ] * **Activate device behavioral fingerprinting -- establish baseline per device type and APN** IOT
- [ ] * **Enable IoT botnet C2 detection -- configure automatic quarantine APN for compromised devices** IOT
- [ ] Configure smart metering security profile -- tamper evidence binding for meter readings IOT
- [ ] Enable GSMA IoT Security Guidelines compliance checks per device category COMPLIANCE
- [ ] Configure IIoT enhanced security profiles for critical infrastructure APNs IOT
- [ ] Verify eSIM/iSIM provisioning integration -- PQC key material in profile download IOT

PHASE DELIVERABLES

- IoT Gateway PQC Activation Report
- Device Behavioral Baseline Report
- Botnet Detection Validation Results
- IoT Security Coverage Report (% devices protected)

PHASE 8

MONITORING & ALERTING

1 week | NOC + QBITEL

Monitoring and alerting configures carrier-grade NOC dashboards, regulatory reporting feeds, and integration with existing SIEM and OSS systems. All KPIs are validated against GSMA fraud reporting standards and operator-specific SLA requirements.

- [ ] * **Configure Prometheus/OpenTelemetry metrics export from all QBITEL Bridge nodes** — MONITORING
- [ ] * **Deploy pre-built Grafana dashboards -- SS7 attacks, fraud events, PQC ops, slice security** — MONITORING
- [ ] * **Configure GSMA fraud KPI reporting -- IRSF block rate, SS7 attack volume, location tracking blocks** — COMPLIANCE
- [ ] Integrate QBITEL Bridge event stream with SIEM (Splunk/Elastic/QRadar) — SIEM
- [ ] Configure PagerDuty/OpsGenie alerting for critical security events — ALERTING
- [ ] * **Enable NIS2 incident detection reporting -- configure 24/72-hour notification workflows** — COMPLIANCE
- [ ] Configure FCC SS7 monitoring monthly report automation — COMPLIANCE

PHASE DELIVERABLES

- NOC Dashboard Deployment and Acceptance
- GSMA KPI Reporting Configuration Validation
- SIEM Integration Test Report
- Alerting Runbook (escalation procedures)

PHASE 9

COMPLIANCE VALIDATION

1 week | Compliance + Legal

Compliance validation generates formal evidence packages for all applicable regulatory frameworks. QBITEL Bridge automated report generation is configured and validated against each framework requirement. Output is a compliance evidence package ready for regulator submission.

- [ ] * **Generate 3GPP TS 33.501 compliance evidence -- SBI mTLS, SUPI protection, slice security controls** — COMPLIANCE
- [ ] * **Generate GSMA FS.19 quantum readiness evidence package -- algorithm inventory, PQC deployment proof** — COMPLIANCE
- [ ] * **Generate GSMA FS.11 SS7 monitoring compliance report -- attack detection and blocking statistics** — COMPLIANCE
- [ ] * **Validate NIS2 Directive compliance evidence -- security measures, incident notification workflows** — COMPLIANCE
- [ ] Generate FCC SS7 remediation evidence report for carrier regulatory filing — COMPLIANCE
- [ ] Conduct NESAS security assurance evidence review with QBITEL Bridge security team — COMPLIANCE
- [ ] * **Legal review of all compliance documentation before submission** — LEGAL
- [ ] * **Obtain operator CISO sign-off on compliance evidence package** — ADMIN

PHASE DELIVERABLES

- 3GPP TS 33.501 Compliance Evidence Package
- GSMA FS.19 Quantum Readiness Certification Evidence
- GSMA FS.11 SS7 Security Compliance Report
- NIS2 Directive Security Measures Documentation
- CISO-Signed Compliance Evidence Package

PHASE
10

GO-LIVE & HANDOVER

3-5 days | Joint QBITEL + Operator

Go-Live confirms full production readiness and transfers day-to-day operational responsibility to the operator NOC and security teams, supported by QBITEL Bridge 24/7 monitoring and ongoing threat intelligence services.

- [ ] * **Conduct final end-to-end SLA validation -- PQC ops/sec, SS7 block latency, availability** SLA
- [ ] * **Execute joint penetration test simulation -- SS7 location tracking, 5G slice bypass attempt** SECURITY
- [ ] * **Complete operator NOC runbook training -- alerting procedures, escalation paths, QBITEL Bridge controls** TRAINING
- [ ] * **Transfer primary operational responsibility to operator NOC with QBITEL Bridge 24/7 monitoring backup** HANDOVER
- [ ] Schedule quarterly security review calendar -- threat intelligence briefings, model updates ONGOING
- [ ] * **Issue Go-Live Certificate and final project documentation package** ADMIN

PHASE DELIVERABLES

- Final SLA Validation Report
- Penetration Test Simulation Results
- Operator NOC Runbook (QBITEL Bridge operations)
- Go-Live Certificate signed by both parties
- Project Closure and Lessons Learned Report

RACI Responsibility Matrix

R = Responsible (does the work) | A = Accountable (signs off) | C = Consulted (provides input) | I = Informed (kept updated)

Activity	QBITEL	Operator NOC	Security Team	Compliance
Network topology and protocol discovery	R	C	I	I
HSM key ceremony	R	I	A	I
SS7 MAP filtering policy definition	C	I	A	C
SS7 inline blocking activation	R	C	A	I
5G SBI sidecar proxy deployment	R	C	I	I
Slice security policy configuration	C	I	A	C
Fraud detection ML model training	R	I	C	I
FMS integration configuration	R	C	C	I
IoT gateway PQC activation	R	C	I	I
NOC dashboard configuration	R	A	C	I
Regulatory compliance evidence generation	R	I	C	A
Go-Live acceptance sign-off	C	I	A	C
Ongoing 24/7 monitoring	R	C	I	I
Quarterly security reviews	R	I	A	C

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